MIL-R-45949A(AT)
23 May 1977
Superseding
MIL-R-45949(WC)
21 February 1973

MILITARY SPECIFICATION

REPAIR KIT, PUNCTURE, PNEUMATIC TIRE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of tubeless tire repair kit which will allow emergency "outside-in" repair of all types and sizes of tubeless heavy-duty tires up to and including 28-ply or ply-rated construction.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-C-96	- Cans, Metal, 28 Gage and Lighter
PPP-C-843	- Cushioning Material, Cellulosic

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Materiel Readiness Command, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 2640

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Military

MIL-P-116	- Preservation - Packaging Method of
MIL-I-3420	- Packaging Materials, Volatile Corrosion Inhibitor Treated, Opaque
MIL-I-8574	- Inhibitor, Corrosion Volatile, Utilization of
MIL-P-14232	- Parts, Equipments and Tools for Army Materiel, Packaging of
MIL-C-16173	- Corrosion Preventive Compound, Solvent Cutback, Cold-Application

STANDARDS

Federal

FED. STD. No. 313	- Symbols for Packages and Containers for Hazardous Industrial Chemicals and Materials
FED. STD. No. 356	- Commercial Packaging of Supplies and Equipment

Military

MIL-STD-129	- Marking for Shipment and Storage

(Copies of specifications, standards, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

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(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois, 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., ATTN: Tariff Order Section, 1616 P Street, N.W. Washington, D.C. 20036.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the supplier shall furnish one complete repair kit for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the supplier of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Design. The repair kit shall be new and a product of the manufacturer's latest design and one that is convenient to operate, has proven successful in operation and is dependable and entirely adequate for the intended purpose. The kit shall be designed to repair punctures in vehicle-mounted tubeless tires of any type or size from the outside in.

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3.3 Construction. The repair kit shall be complete so that it can be used for the operations specified herein. All surfaces of all parts shall be clean and free of extraneous materials. External surfaces shall be smooth and free of sharp edges. All screws,pins and similar parts as might be used shall be installed with means to prevent loss of tightness or fit. All such parts subject to adjustment or removal shall not be swaged, peened, staked or otherwise permanently deformed.

3.3.1 Materials. All parts and materials used in the fabrication of the repair kit shall be new and of a quality necessary to produce a kit that will meet the requirements specified herein. If in the construction of kit, widely dissimilar metals are used, especially brass, copper or steel, contact with aluminum or magnesium shall be avoided wherever possible.

3.3.2 Interchangeability. All replaceable parts shall, wherever possible be manufactured to such standards as will permit interchangeability, replacement or adjustment without modification of parts.

3.3.3 Castings and Forgings. All castings and forgings where used shall be free from defects. Strength and other essential physical and chemical properties of any castings or forgings shall be adequate to meet performance requirements specified herein.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering where used shall be employed only when specified in the original design and shall be of good commercial quality.

3.3.5 Painting. Unless otherwise specified (see 6.2), painting and finishing of the repair kit container and components shall be in accordance with the supplier's standard commercial practice provided the following minimum criteria are met or exceeded: all surfaces to be painted shall have been cleaned to be free of all foreign matter detrimental to painting, and at least one coat of primer and one coat of enamel or equivalent have been applied. Also final topcoat color shall be either dark green or olive drab.

3.3.6 Threads. Where threaded parts are used, the thread shall be in the inch system and comply with Handbook H28.

3.4 Performance and product characteristics.

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3.4.1 Performance. The repair kit shall be capable of repairing punctures ranging from nail holes to slits five (5) inches in length, and holes up to 0.75 inch in diameter in all types and sizes of tubeless tire casings up to and including 28-ply or ply-rated construction.

3.4.2 Kit components. Unless otherwise specified (see 6.2.1), the repair kit shall consist of the following principal components in the quantities indicated or as specified:

(a) Insertion tool, handle type - one each.

(b) Refills, repair material - four each.

(c) Bonding Compound - two each.

(d) Leak detector material - one each.

(e) Leak detector applicator bottle - one each.

(f) Skiving knife - one each.

(g) Reamer Tool - one each.

(h) Metal Box - one each.

3.4.3 Detail of components.

3.4.3.1 Insertion tool. The insertion tool shall consist of an insertion needle with either a permanently attached or detachable handle. The handle shall be either solid or hollow tubular in construction with opposite end of handle from needle end permanently closed and free from burrs and sharp edges. It shall have a diameter of $1\frac{1}{2}$ inches $\pm$ $\frac{1}{2}$ inch and length of $6\frac{1}{4}$ inches $\pm$ $\frac{1}{2}$ inch. Material and permanent or mechanical assembly shall be optional but capable of retaining insertion needle when in use. The needle shall be made of corrosion resistant steel, plated carbon steel or plated alloy steel, and be of sufficient diameter strength and hardness as to be suitable for its intended purpose. It shall have a suitable eye to accommodate the repair material to be used for insertion into injury of tubeless tire. The needle shall when mounted in handle have an effective usable length of from $1\frac{1}{2}$ to 8 inches if adjustable and 8 inches if permanently mounted. It shall be capable of withstanding a minimum longitudinal force of 100 pounds (see 4.4.2) and perform its intended function (see 3.4.1) without breaking or being permanently deformed.

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3.4.3.2 Refills, repair material. The refills of repair material shall be a minimum of 600 inches of sealant material consisting of a polypropylene or nylon sleeve impregnated and coated with a polysulfide rubber compound and having a minimum tensile strength of 5,100 pounds per square inch (psi). The repair material shall be round or elliptical in shape with a cross sectional diameter of not less than 0.090 inch to not more than 0.110 inch. The repair material shall be coiled and encapsulated in a plastic tube properly identified as to the material and length inside. Provision for capping ends of tube shall be provided to retain repair material in tube when not in use.

3.4.3.3 Bonding compound. The bonding compound material shall be of styrenebutadiene or natural rubber with aliphatic hydrocarbons and solid content of 19 to 21 percent or with toluene and a solid content of 14 to 16 percent. The bonding compound shall be furnished in two (2) eight (8) ounce, screw-top cans. The cans shall be properly identified as to content and have reasonably unlimited storage life under normal conditions.

3.4.3.4 Leak detector. The leak detector material shall be in the form of water soluble concentrated liquid. When properly mixed with water, it shall have the capacity of detecting and showing leakage or seepage of pressure from a tubeless tire. A minimum of two (2) fluid ounces of leak detector material shall be furnished in each kit. It shall be contained in a screw-cap-type plastic container which shall be leak-proof and capable of preventing moisture absorption from the air. It shall under normal conditions have a reasonably unlimited storage life and be identified as to its contents and capacity.

3.4.3.5 Leak detector applicator bottle. The leak detector bottle shall be of plastic, four to eight (4 to 8) ounce capacity, squeeze dispensing-type for mixing the leak detector material with water before use and application of mixed solution to tubeless tire to allow visual detection of any leak or seepage of pressure. The applicator bottle shall incorporate a screw-type applicator cap for filling and dispensing and an attached cap for applicator tip. It shall also be properly identified as to its use.

3.4.3.6 Skiving Knife. A skiving knife with a 4 to 6½ inch blade shall be provided. It shall have a square point and be of high quality steel permanently mounted in either a metal, wood or plastic handle.

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3.4.3.7 Reamer Tool. The reamer tool shall consist of a round tapered reamer needle type instrument permanently mounted in either a metal, wood or plastic screwdriver type handle. The handle shall be approximately $1\frac{1}{2}$ inches in diameter $\pm\frac{1}{4}$ inch with a minimum length of 3 inches and a maximum length of 5 inches. The reamer portion of tool shall be of carbon or alloy steel hardened to Rockwell C 44-55. It shall have a tapering diameter starting with a point and tapering out to 0.180 to 0.200 inches diameter in a working usable length of 4 inches $\pm\frac{1}{2}$ inch at the handle. The latter two thirds of the reamer measured from the handle shall be serrated, serrations having the ability to cut and enlarge punctures in tire casings of any type or size. Tool must be capable of performing the described intended function without dislodging handle, bending, breaking or permanently deforming reamer portion of tool in anyway (see 4.4.3).

3.4.3.8 Metal box. The box shall be fabricated of not less than 24 gage metal and be rectangular in shape with all sides flat including the lid. The lid shall be piano-hinged either full length or a minimum of two (2) smaller ones located toward ends of lid. The box and lid shall have a positive latching device with an instruction data plate permanently affixed to the inside of the lid. It shall be capable of containing all component parts of kit and shall be painted in accordance with manufacturer's standard commercial practice (see 3.3.5).

3.5 Identification marking. The repair kit shall be marked for identification in accordance with MIL-STD-130 and unless otherwise specified (see 6.2.1) shall include the National Stock Number.

3.6 Technical data. Data shall be furnished as specified (see 6.2.1 and 6.2.2).

3.7 Workmanship. The repair kit shall be processed in such a manner as to be free from cracked or misplaced parts, sharp edges, burrs and other defects which will affect life, serviceability or appearance.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the

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the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2.1)

(b) Quality conformance inspection (see 4.2.2)

4.2.1 First article inspection. First article inspection shall be applied to the preproduction model or initial production item as might be noted in 3.1. Unless otherwise specified (see 6.2.1) first article inspection shall consist of the examination in 4.3 and all tests under 4.4. Failure of the first article to pass any examination or test shall be cause for rejection.

4.2.2 Quality conformance inspection sampling. Quality conformance inspection sampling shall be applied to production items offered for acceptance under the contract. As a minimum, quality conformance inspection shall consist of (a) through (c) as follows:

(a) Examination (see 4.3)

(b) Certification requirements (see 4.3.1)

(c) Preparation for delivery inspection (see 4.5)

4.3 Examination. Each repair kit shall be visually examined to determine compliance with requirements of section 3, using the following list:

Examination Check List	Requirement Reference
Painting	3.3.5
Kit components	3.4.2
Insertion Tool	3.4.3.1
Refills, Repair Material	3.4.3.2
Bonding compound	3.4.3.3
Leak detector material	3.4.3.4
Leak detector applicator bottle	3.4.3.5

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Examination Check List	Requirement Reference
Skiving knife	3.4.3.6
Reamer tool	3.4.3.7
Metal box	3.4.3.8
Identification marking	3.5

4.3.1 Certification requirements. A certified test report covering items and requirements listed in 4.4.1 shall be maintained by the contractor and shall be available for Government inspection.

4.4 Tests.

4.4.1 The characteristics of the following components shall be tested to verify compliance with the requirements of section 3:

	Requirement
(a) Refill, repair material	
Identity of core	Nylon or polypropylene
Linear length, inches	600 min
Tensile strength, psi	5100 min
Cross sectional diameter, range, inches	0.090 to 0.110
(b) Bonding compound	
Identity:	
Solid	Natural rubber or sytrenebutadiene rubber
Solvent	Aliphatic hydrocarbons
Solids content, %	19 - 21
Solvent	Toluene
Solids content, %	14 - 16
(c) Leak detector	
Identity:	Water soluble concentrated liquid
(d) Needle, Insertion tool	
Material	Corrosion resistant steel, plated carbon steel or plated alloy alloy

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(e) Reamer tool	Requirement
Material	Carbon or Alloy steel, Rockwell hardness C44-55

4.4.2 Force test, Insertion tool. With handle affixed to insertion needle and needle protruding from handle a minimum of 1½ to a maximum of eight (8) inches from handle, a longitudinal force of 100 pounds shall be applied to the eye end of the needle. The needle shall not move in or dislodge from handle nor shall the needle break or be permanently deformed (see 3.4.3.1).

4.4.3 Force test, Reamer tool. The reamer tool shall have the capability of being forced into, through and enlarging the average nail type puncture at the thickest or stiffest portion of all types and sizes of tubeless tire casings up to and including 28-ply or ply-rated construction. Tool shall within a reasonable life span perform all of the above requirements without failure or damage of any kind (see 3.4.3.7 and 4.4.1).

4.5 Preparation for delivery inspection. The preservation, packaging, packing and marking of the repair kits shall be inspected to determine compliance with the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or B or Commercial as specified (see 6.2.1),

5.1.1 Level A.

5.1.1.1 Cleaning. Components of kit shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Components of kit shall be dried in accordance with the applicable procedures of MIL-P-116.

5.1.1.3 Preservation and Packaging. Unit method of protection for repair kit shall conform to packaging method 1 of MIL-P-116. Immediately after cleaning and drying, a continuous coating of preservative MIL-C-16173 Grade 1 or Grade 4 shall be applied to the metallic surface of knife and exposed section of insertion needle and related exterior bare ferrous metal parts of insertion tool. The interior surface of insertion tool shall be preserved with volatile corrosion inhibitor (VCI) (P-18) MIL-I-3420. The VCI shall be used and applied in accordance with

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MIL-I-8574. The bonding compound shall be sealed in two (2) metal cans conforming to PPP-C-96, Type VIII with screw cap closure. Repair material shall be housed in a plastic cartridge. The leak detector concentrate shall be sealed in a plastic bottle or a plastic jar with screw cap closure. Place components of kit in kit container as necessary and immobilize with cushioning material conforming to PPP-C-843.

5.1.2 Level B. Cleaning, drying, preservation and packaging shall be the same as Level A.

5.1.3 Commercial Packaging. Cleaning, drying, preservation, packaging and marking shall be in accordance with Fed Std No 356.

5.2 Packing. Packing shall be Level A, B or Commercial (see 6.2.1).

5.2.1 Level A. Packaged repair kits shall be packed into exterior shipping containers, conforming to the applicable Level A requirements of MIL-P-14232.

5.2.2 Level B. Packaged repair kits shall be packed into exterior shipping containers conforming to the applicable requirements of MIL-P-14232.

5.2.3 Commercial. Packing shall conform to the requirements of Fed Std No 356.

5.3 Marking. In addition to any special marking requirements specified by the contract or purchase order, unit packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The tire repair kit covered by this specification is intended for field use on tactical vehicles utilizing tubeless heavy-duty tires up to and including 28-ply or ply-rated construction. The kit shall allow a temporary "get-home" capability of 40 miles minimum. Repair to the tire shall be performed "on-the-wheel".

6.2 Ordering data. Procurement documents should specify:

6.2.1 Procurement requirements.

(a) Title, number, and date of this specification.

(b) First article, when required (see 3.1).

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(c) Painting, if different (see 3.3.5).

(d) Kit components, if different (see 3.4.2).

(e) Identification marking, if different (see 3.5).

(f) First article inspection, if different (see 4.2.1).

(g) Material Safety Data Sheet shall be furnished for each contract (see 3.6, 6.2.1 and 6.2.2).

(h) Selection of applicable level of preservation and packaging (see 5.1).

(i) Selection of applicable level of packing (see 5.2).

6.2.2 Contract data requirements. Required technical data, such as operators manuals, parts lists, assembly directions, and other instruction for operation and maintenance, as identified on a DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract (see 3.6).

Custodian:
Army - AT

Review activity:
Army - ME, SM

User activity:
Army - GL

Preparing Activity:
Army - AT

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DOCUMENT IDENTIFIER AND TITLE
MIL-R-45949A(AT)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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